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NATURAL GAS

Physical markets & trading



From the wellhead to a trading decision

The physical system determines which trades are possible; contracts determine who earns the margin.

01
PRODUCTION
Associated or non-associated gas
Well output

→

02
PROCESSING
Separate NGLs and impurities
Pipeline specification

→

03
TRANSPORT + STORAGE
Move gas across location and time
Capacity and rights

→

04
CUSTOMER
Power, industry, buildings, LNG
Delivered demand

MEETING ROUTE

01 MARKET FUNDAMENTALS
Demand, production, infrastructure and storage
Slides 3–9

02 ACCESS + PRICING
Basis, transport costs and service rights
Slides 10–11

03 MERCHANT TRADING
Arbitrage, customer margins and risk
Slides 12–18

A price is not enough: every physical trade needs a **location, delivery period, volume and access rights.**

Source: supplied CTC Natural Gas study guide and research.

DEMAND FUNDAMENTALS

Natural gas is both a fuel and a manufacturing input

Mostly methane (CH_4): power, industry and buildings drive consumption; transportation is a smaller use.

01	**ELECTRICITY**	Gas-fired plants generate power and can adjust output relatively quickly as demand changes.
02	**INDUSTRY**	High-temperature heat for glass, steel and cement; feedstock for chemicals and ammonia fertilizer.
03	**BUILDINGS**	Space heating, hot water and cooking in homes and businesses. Cold weather lifts demand.
04	**TRANSPORT**	Some trucks, buses and ships use gas, but this remains smaller than power, industry and heating.



LNG • TRANSPORT BY SHIP

Gas cooled to approximately **−162°C** becomes liquid, allowing trade without a direct pipeline.

DO NOT CONFUSE

LNG = liquefied natural gas
A transport form of methane

NGLs = natural gas liquids
Ethane, propane and butane separated during processing

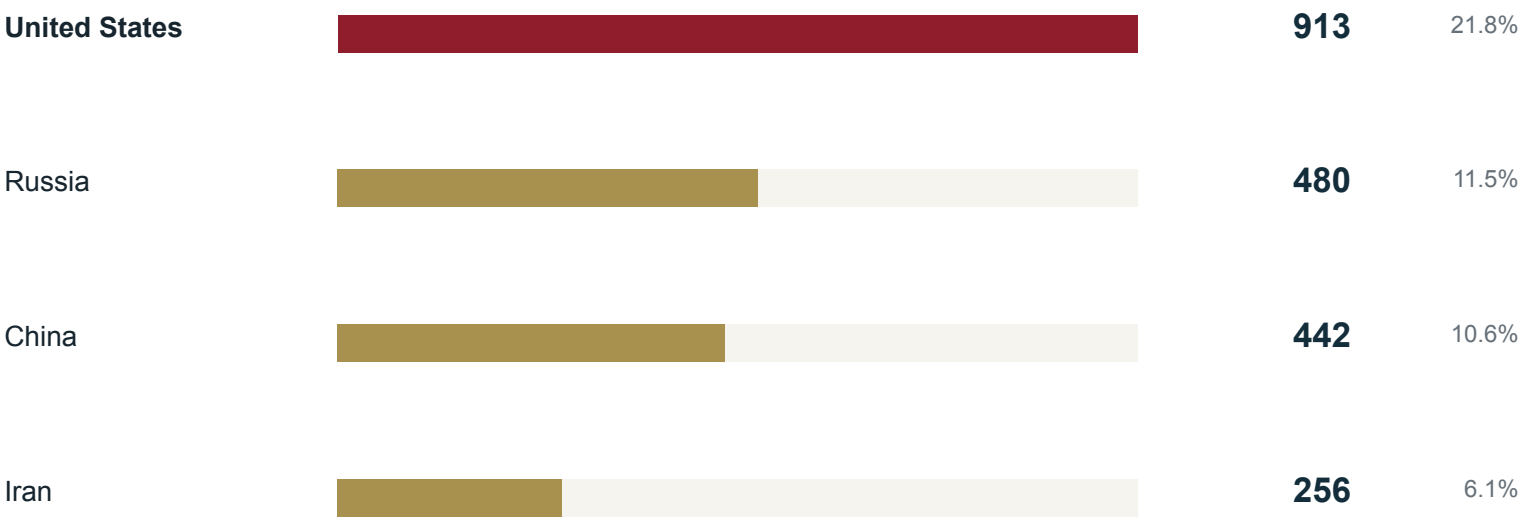
Source: supplied CTC Natural Gas study guide and research. LNG carrier image: Dreamstime illustration.

DEMAND FUNDAMENTALS

Four countries account for roughly half of global demand

2025 statistics reproduced from the supplied material; electricity is the largest single use highlighted.

2025 CONSUMPTION • BCM AND WORLD SHARE



Top four: 913 + 480 + 442 + 256 = 2,091 bcm, approximately **50%** of reported global demand.

GLOBAL CONSUMPTION

4.2 tcm

reported for 2025

ELECTRICITY SHARE

≈34%

U.S. DOMESTIC CONSUMPTION

33.5 Tcf

2025 • approximately 92 Bcf/day
LNG exports: 15.1 Bcf/day

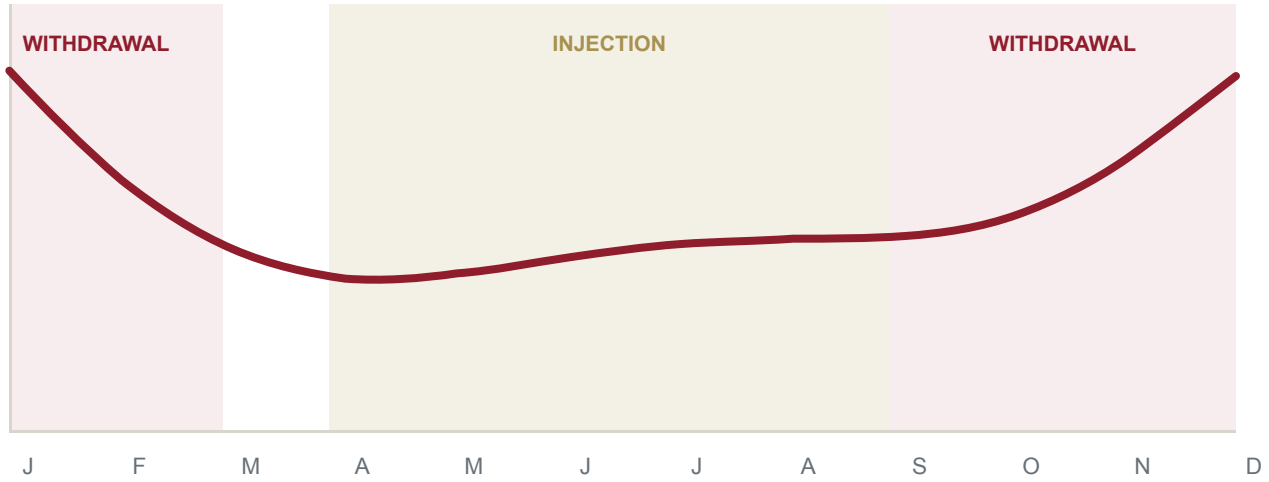
Source: supplied research citing International Gas Union and EIA. Historical/source-reproduced figures, not live data.

DEMAND FUNDAMENTALS

Weather creates the largest cyclical short-term demand swings

Production is relatively less seasonal; storage and prices help balance winter heating and summer power demand.

TYPICAL U.S. DEMAND PATTERN • SCHEMATIC



Typical injection: **April–October**
Typical withdrawal: **November–March**

FIVE DEMAND FACTORS TO WATCH

01 **Weather**
Heating and cooling needs

02 **Electricity generation**
Gas plant dispatch and competing generation

03 **Industrial production**
Factory activity and feedstock demand

04 **International LNG demand**
Exports connect U.S. supply to global buyers

05 **Prices**
Fuel switching and demand response

January 2025: **126.6 Bcf/day**

versus approximately **92 Bcf/day** for full-year 2025

Source: supplied CTC Natural Gas study guide and research; seasonal curve is schematic and not to scale.

PRODUCTION + INFRASTRUCTURE

Natural gas supply chain

Gas must meet pipeline specifications and have capacity to reach the buyer.

01

GATHERING

Collects gas from individual wells

Small, low-pressure lines

Producers / midstream

02

PROCESSING

Separates NGLs and impurities

Plant; pipeline specification

Midstream

03

TRANSMISSION

Moves gas from basin to market

Large, high-pressure pipelines

Pipeline operators

04

DISTRIBUTION

Supplies homes and businesses

Small, low-pressure local system

Local utilities (LDCs)

PHYSICAL CONSTRAINTS

PRESSURE

Gas flows from higher to lower pressure.

CAPACITY

Diameter and compression constrain daily throughput.

PIPELINE FUEL

100 units in → roughly 97–99 delivered.

Production alone does not guarantee **deliverable supply** at the market that needs it.

Source: supplied CTC Natural Gas study guide and research.

PRODUCTION + INFRASTRUCTURE

Supply responds to different economics—and different processing needs

Associated gas follows oil drilling; non-associated gas is drilled primarily for gas.

PRODUCTION ECONOMICS

Associated gas

Produced alongside oil. Supply is strongly influenced by **oil economics** and drilling decisions.

Non-associated gas

Wells drilled primarily for gas. Supply responds more directly to **gas economics**.

GAS TYPE

Wet / rich gas

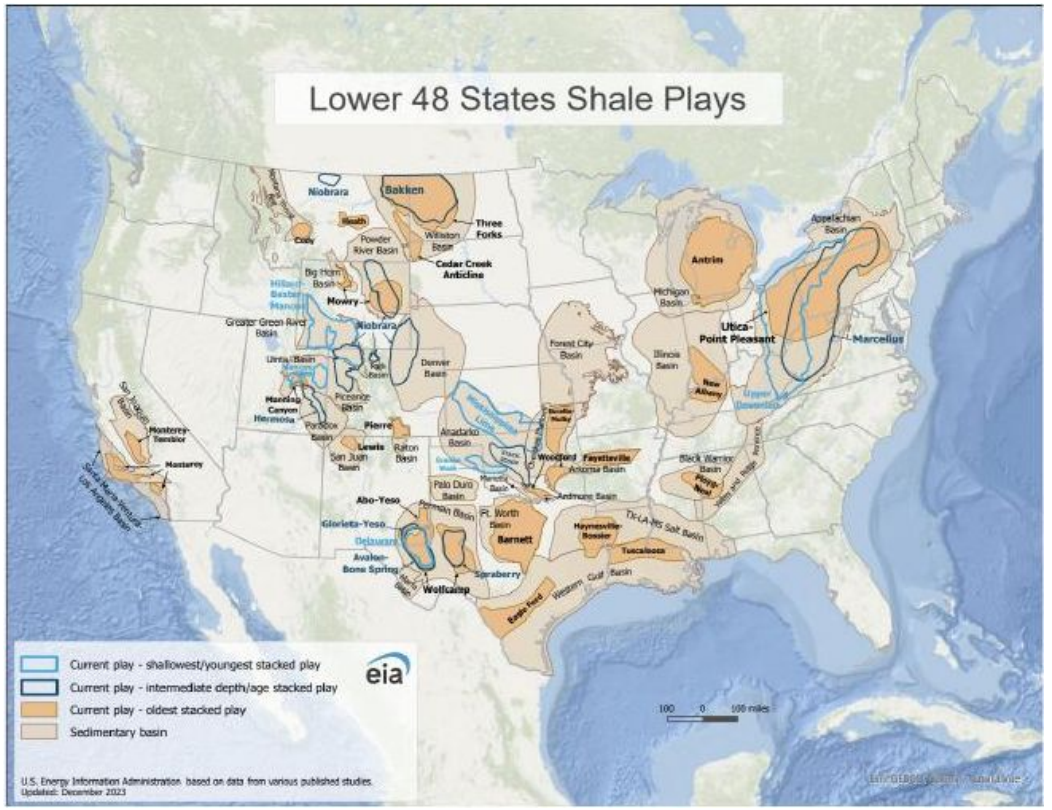
More NGLs; generally needs **more processing** before long-haul transmission.

Dry / lean gas

Mostly methane; generally needs **less processing** before transmission.

PERMIAN CORRECTION

Gas produced as a byproduct of oil drilling is **associated gas**. Weak gas prices do not automatically stop supply if oil drilling remains attractive.



Appalachia: Marcellus and Utica formations

Source: supplied study guide and research; EIA Lower 48 shale-play map reproduced there (updated Dec. 2021).

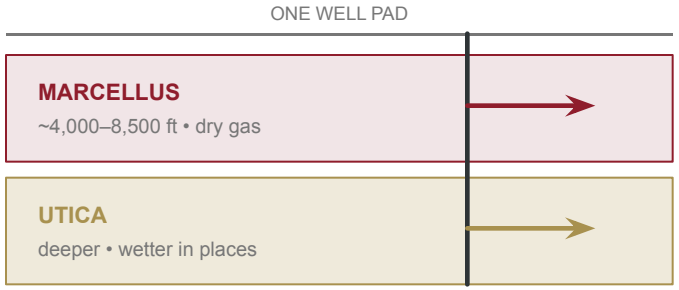
PRODUCTION + INFRASTRUCTURE

Three stacked plays dominate U.S. supply; pipelines decide what it earns

Plays are the geology, the stack is the cost advantage, and takeaway capacity turns location into basis.

THE STACK • PAY ZONES IN LAYERS

APPALACHIA CROSS-SECTION • SCHEMATIC, NOT TO SCALE



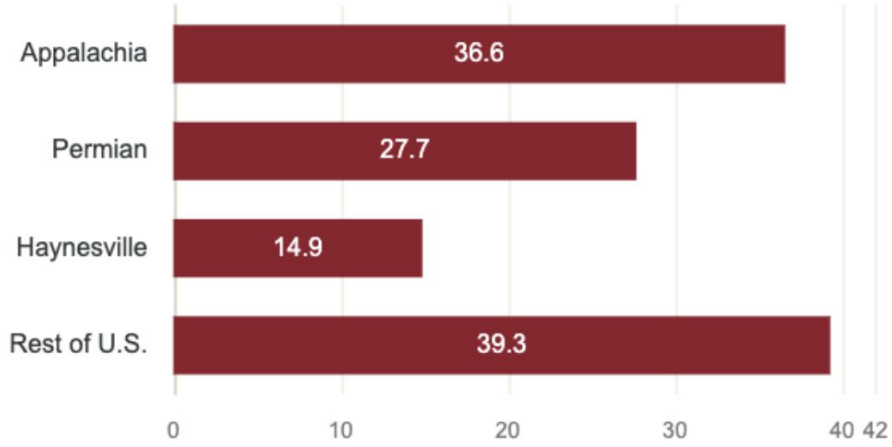
One vertical wellbore can land horizontal laterals in more than one pay zone.

- Stacked pay = several productive zones under one surface location
- Appalachia: Marcellus over Utica • Permian: multiple benches (Spraberry, Bone Spring, Wolfcamp)
- One pad, road and gathering system spread over more reserves per acre

DEPTH TRADE-OFF — Haynesville wells sit deeper (~10,500–13,500 ft) and cost more to drill; proximity to Gulf Coast LNG demand offsets it.

MAIN PRODUCTION AREAS • 2025

MARKETED PRODUCTION • BCF/D, ANNUAL AVERAGE



67% of U.S. output (118.5 Bcf/d record) — and **81%** of 2025 growth — came from these three regions.

Appalachia = Marcellus + Utica, dry gas • **Permian** = associated, follows oil economics • **Haynesville** = dry gas beside Gulf LNG demand

Geology determines how much gas exists; pipeline capacity determines where it earns — basis is the scoreboard.

Source: EIA, Today in Energy (Mar 2026) and Natural Gas Monthly — 2025 annual-average marketed production; EIA well-depth comparisons. Cross-section is schematic.

PRICING + TRANSPORTATION

Pipelines set the basis — the price of location

Use the same delivery period and compatible terms: basis = local price − benchmark price.

BASIS = $P_{local} - P_{benchmark}$

HENRY HUB

Physical pipeline hub in **Erath, Louisiana**. It serves as the pricing benchmark for North American natural gas futures.

READING THE SIGN

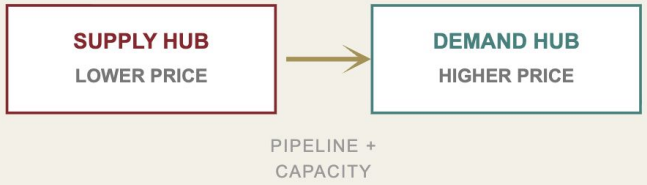
Positive basis: local > benchmark

Negative basis: local < benchmark

Supply hubs clear below the benchmark when pipes are full; demand hubs clear above it on cold days.

WHY BASIS MOVES

- Regional supply
- Regional demand
- Transmission bottlenecks
- Delivery period and terms



The spread between them is the gross margin for whoever holds capacity.

PIPELINES SET THE BASIS

01 **APPALACHIA → SOUTHEAST**
Mountain Valley, 2.0 Bcf/d, in service Jun 2024 — takeaway limits capped growth for years

02 **PERMIAN (WAHA) → GULF COAST**
Matterhorn, 2.5 Bcf/d, in service Sep 2024 — with pipes full, Waha traded below $0 on 46% of 2024 days

03 **HAYNESVILLE → LNG CORRIDOR**
Short haul to Gulf Coast liquefaction — the basin grows with export demand

FULL PIPE = WIDER DISCOUNT — new capacity narrows the basis; that spread is the trading opportunity.

A regional spread is not yet a profit: subtract variable transport, pipeline fuel and allocated reservation costs.

Source: supplied CTC Natural Gas study guide and research; EIA pipeline-project notes (Sep 2024). Diagram is conceptual; no live prices shown.

STORAGE

Storage balances seasons—but deliverability matters on the coldest day

Capacity is how much gas can be held; injection and withdrawal rates determine how quickly it can move.

01 DEPLETED RESERVOIRS

Reused oil and gas fields; known geology can reduce development cost.

Usually slower, seasonal cycling

Largest share of U.S. capacity in the notes

02 SALT CAVERNS

Cavities formed in salt, especially near the Gulf Coast.

Fast injection and withdrawal; multiple cycles

Cold snaps, price spikes and LNG feed gas swings

03 AQUIFERS

Water-bearing formations adapted for storage, especially in the Midwest.

Higher cost and less flexibility

Large base-gas requirement

THE SEASONAL ROLE

APR–OCT • INJECTION	NOV–MAR • WITHDRAWAL

Typical seasons, not restrictions on daily operations. Storage absorbs the difference between steadier supply and seasonal demand.

INVENTORY ≠ IMMEDIATE SUPPLY

A cold-snap trade needs sufficient **withdrawal rate and reliable rights**—not just gas in the ground.

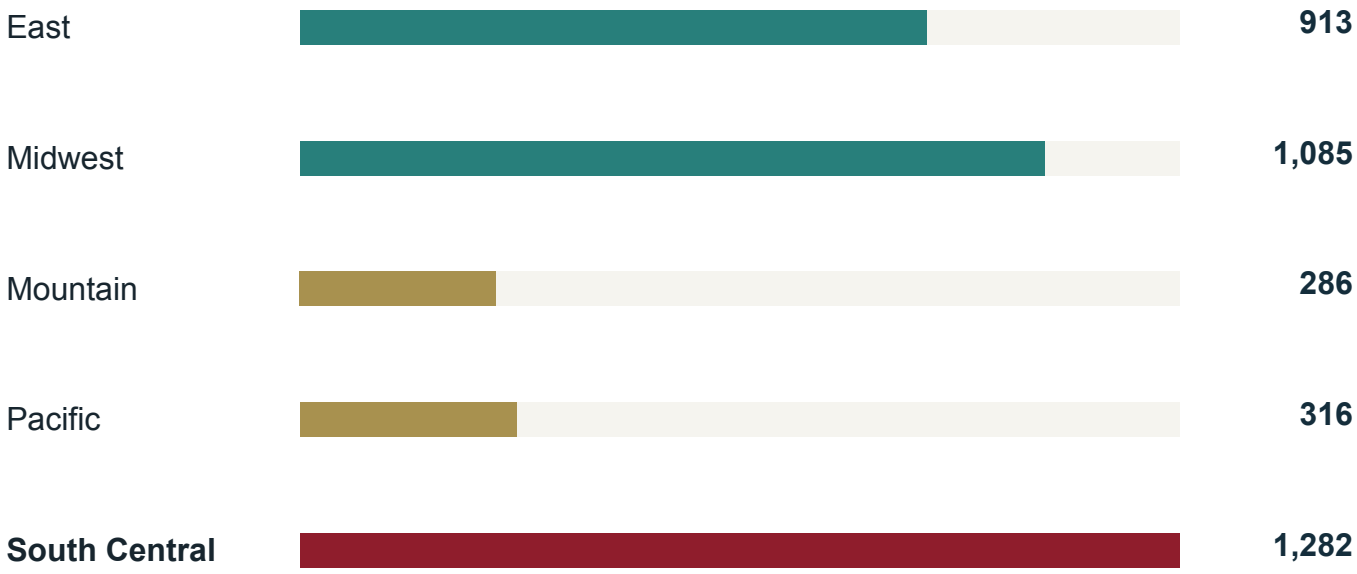
Source: supplied CTC Natural Gas study guide and research.

STORAGE

Where storage sits changes its trading value

Working gas in underground storage, Lower 48 states — 24 October 2025 snapshot.

WORKING GAS • BCF



TOTAL WORKING GAS 3,882 Bcf

≈ 51%

EAST + MIDWEST

Near winter consumers; mostly depleted reservoirs and aquifers. Local supply helps meet heating peaks.

≈ 33%

SOUTH CENTRAL

Near production, industry, LNG exports and Henry Hub; Gulf salt adds fast flexibility.

New England: little suitable underground storage; pipelines, LNG and peak-shaving tanks contribute to the Algonquin winter premium.

Source: supplied study guide, EIA working-gas snapshot dated 24 Oct 2025. Shares are of working gas, not installed capacity.

PRICING + TRANSPORTATION

Firm reservation cost: Guaranteed right

Reservation charges buy priority; interruptible service can be curtailed on peak days.

SERVICE	WHAT YOU PAY	WHAT YOU RESERVE	PEAK-DAY REALITY
Firm transport	**Reservation charge, used or not**; plus usage charges	Pipeline capacity	Reserved priority
Interruptible transport	Mainly when used	No reserved priority	Can be curtailed
Firm storage	Reservation + **injection / withdrawal fees**	Space + specified rates	Reliable deliverability
Interruptible storage	Charges on actual service; no firm reservation	Spare capacity only	Injection or withdrawal may be restricted

CAPACITY RELEASE

A firm-capacity holder resells rights it does not plan to use.

AMA

A merchant manages a utility's capacity, meets supply obligations and trades spare capacity.

COLD-SNAP RULE: prioritize firm deliverability—not merely cheap space.

Source: supplied CTC Natural Gas study guide and research. Exact rights and charges depend on the contract.

MERCHANT STRATEGIES

Location arbitrage: one cold day can justify holding capacity

Illustrative Appalachia → New York trade; firm capacity of 50,000 MMBtu/day.

BUY • APPALACHIA	SELL • NEW YORK	VARIABLE COST	ALLOCATED RESERVATION
$2.50	$6.00	$0.10	$0.40

COLD-DAY CALCULATION

($6.00 − $2.50 − $0.10 − $0.40) × 50,000

$150,000

daily profit • $3.00/MMBtu

MILD DAY: FLOW OR STAY IDLE?

Buy $2.00 • Sell $2.20 • Spread $0.20

FLOW	IDLE
−$15,000	−$20,000

THE DAILY DECISION USES VARIABLE COST

The reservation fee is already committed. Flow if the spread exceeds variable cost; on the mild day, flowing loses less than idling.

Annual economics must still cover reservation charges.

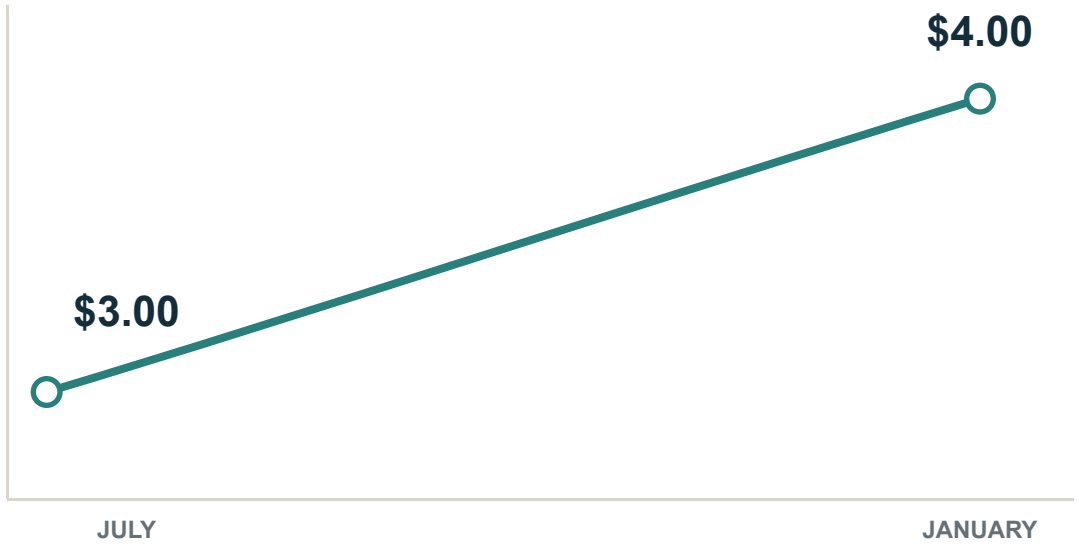
Source: supplied study guide. Illustrative prices and simplified costs; assumes capacity, delivery and operating conditions are achieved.

MERCHANT STRATEGIES

Time arbitrage: store gas only when the spread covers all costs

Illustrative July-to-January storage carry; 1 Bcf ≈ 1 million MMBtu.

CONTANGO • LATER DELIVERY COSTS MORE



Buy and inject in July; sell January futures at the same time. Withdraw and deliver in January.

LOCKING THE STORAGE CARRY

01 BUY + INJECT

1 million MMBtu at $3.00

02 SELL FORWARD

January futures at $4.00

03 PAY ALL STORAGE COSTS

$0.50/MMBtu total stated cost

($4.00 − $3.00 − $0.50) × 1,000,000

$500,000 • illustrative profit

Intrinsic = locked in at the start. **Extrinsic** = later optimization; include hedge-close and physical-plan costs.

Source: supplied study guide. Illustrative prices and simplified storage costs; futures prices are not a forecast of future spot prices.

MERCHANT STRATEGIES

Merchants also earn margins by solving customer problems

Asset optimization, origination and hedging use contracts and risk offsets—not just directional price views.

01 ASSET OPTIMIZATION

Boston utility AMA

SIMPLIFIED CALCULATION

$1.8M trading gains
− $1.5M utility fee

$300,000

Merchant prioritizes utility supply, then trades spare Appalachia → Algonquin capacity.

02 ORIGINATION

Fixed-price factory supply

SIMPLIFIED CALCULATION

($4.20 − $3.80 − $0.25)
× 1.8 million MMBtu

$270,000

Back-to-back futures and transport support the customer's delivered fixed price.

03 RISK INTERMEDIATION

Haynesville producer hedge

SIMPLIFIED CALCULATION

($3.50 − $3.45)
× 18.25 million MMBtu

$912,500

A custom fixed-price swap is offset with futures; the spread pays for customization.

The merchant earns a margin for handling **assets, delivery, flexibility and customized risk**.

Source: supplied study guide. All examples are illustrative and simplify matched exposures, delivery and contract performance.

TRADING RISKS

Unexpected volume can erase a hedge's margin

The factory's fixed-price deal assumed 20,000 MMBtu/day; flexible demand creates an unhedged option.

COLD-SNAP VOLUME EXAMPLE

Factory takes an **extra 5,000 MMBtu/day**. The merchant buys replacement gas in the cash market but still sells at the fixed contract price.

CASH PURCHASE	FIXED SALE	EXTRA VOLUME
$7.00	$4.20	5,000/day

($4.20 − $7.00) × 5,000 = −$14,000/day

Loss of $2.80/MMBtu before additional costs

ORIGINAL $270,000 MARGIN LASTS

19.3 days

at a $14,000 daily loss on the extra volume

Flexible volume is a customer option—it must be priced and risk-managed.

OPERATIONAL

Scheduling errors, imbalances and service interruptions

CREDIT

Customer or supplier fails to pay or deliver

VOLUME + WEATHER

Demand changes when replacement gas is expensive

Source: supplied study guide. Illustrative risk example; $270,000 ÷ $14,000/day ≈ 19.3 days, before extra costs.

INTERACTIVE EXERCISE

Boston winter: what would you do with the spare capacity?

Practice scenario • illustrative inputs • teams of 3–4 • 6 minutes + 4-minute debrief

YOUR POSITION

30,000 MMBtu/day firm Appalachia → Algonquin capacity. The utility needs **20,000**, leaving **10,000 spare**.

COLD-DAY MARKET

Buy **$2.50** Sell **$6.00**

Variable cost **$0.10**

Allocated reservation **$0.40**

SEPARATE CUSTOMER REQUEST

Factory wants **5,000 MMBtu/day extra** at $4.20 while replacement gas costs $6.00.

YOUR TEAM HAS FOUR DECISIONS

01 **FLOW OR IDLE?**

Should you use the 10,000 MMBtu/day of spare capacity?

02 **CALCULATE THE TRADE**

Show net $/MMBtu and total daily profit after all stated costs.

03 **PRICE THE FLEXIBILITY**

Accept the factory request, reprice it, or decline? Show why.

04 **CHECK THE PHYSICS + RISK**

Which rights must be firm? Name two things that could still break.

NEXT: TEAM WORKSHEET → record decisions before the answer reveal

Source: practice scenario constructed from the supplied location-arbitrage and factory volume-risk examples. Not live market data.

EXERCISE DEBRIEF • ANSWER REVEAL

Flow the spare capacity—but do not give away flexibility

Click to reveal each answer during the debrief; all figures are illustrative.

01 **FLOW • $3.50 SPREAD > $0.10 VARIABLE COST**

The reservation fee is sunk for the immediate flow decision; include it when calculating full profitability.

02 **$3.00/MMBtu → $30,000/DAY**

($6.00 − $2.50 − $0.10 − $0.40) × 10,000 = $30,000

03 **REPRICE OR DECLINE • −$14,000/DAY AT OLD TERMS**

Replacement: ($4.20 − $6.00) × 5,000 = −$9,000 before extra costs. Flexible volume is a customer option.
Transport: ($4.20 - $2.5 - $0.10 - $0.40) × 5000 = $6,000

04 **FIRM TRANSPORT + RELIABLE DELIVERABILITY**

Check nominations, interruptions, scheduling and credit. If storage is used, secure firm withdrawal rights.

STRONG ANSWERS SEPARATE price spread • physical access • contract exposure

Source: suggested solution constructed from the supplied location-arbitrage and factory volume-risk examples.

TRANSFER + DISCUSSION

Five questions to ask before approving a physical gas trade

Use this checklist to connect the commodity, the asset and the contract.

01	**WHAT EXACTLY ARE WE BUYING AND SELLING?**	Hub, delivery period, volume and MMBtu terms
02	**CAN THE GAS PHYSICALLY MOVE?**	Processing, route, pipeline capacity and compressor fuel
03	**CAN INVENTORY DELIVER WHEN NEEDED?**	Working volume, injection / withdrawal rates and firm rights
04	**IS THE MARGIN POSITIVE AFTER EVERY COST?**	Reservation, variable charges, storage, fuel and hedge-close costs
05	**WHAT CAN BREAK THE PLAN?**	Weather, unexpected volume, operations and counterparty credit

Physical merchants combine **assets, flow information and scale**. A price view alone is not a trading strategy.

ANY QUESTIONS?

Source: synthesis of the supplied CTC Natural Gas study guide. All numerical examples in this deck are illustrative.